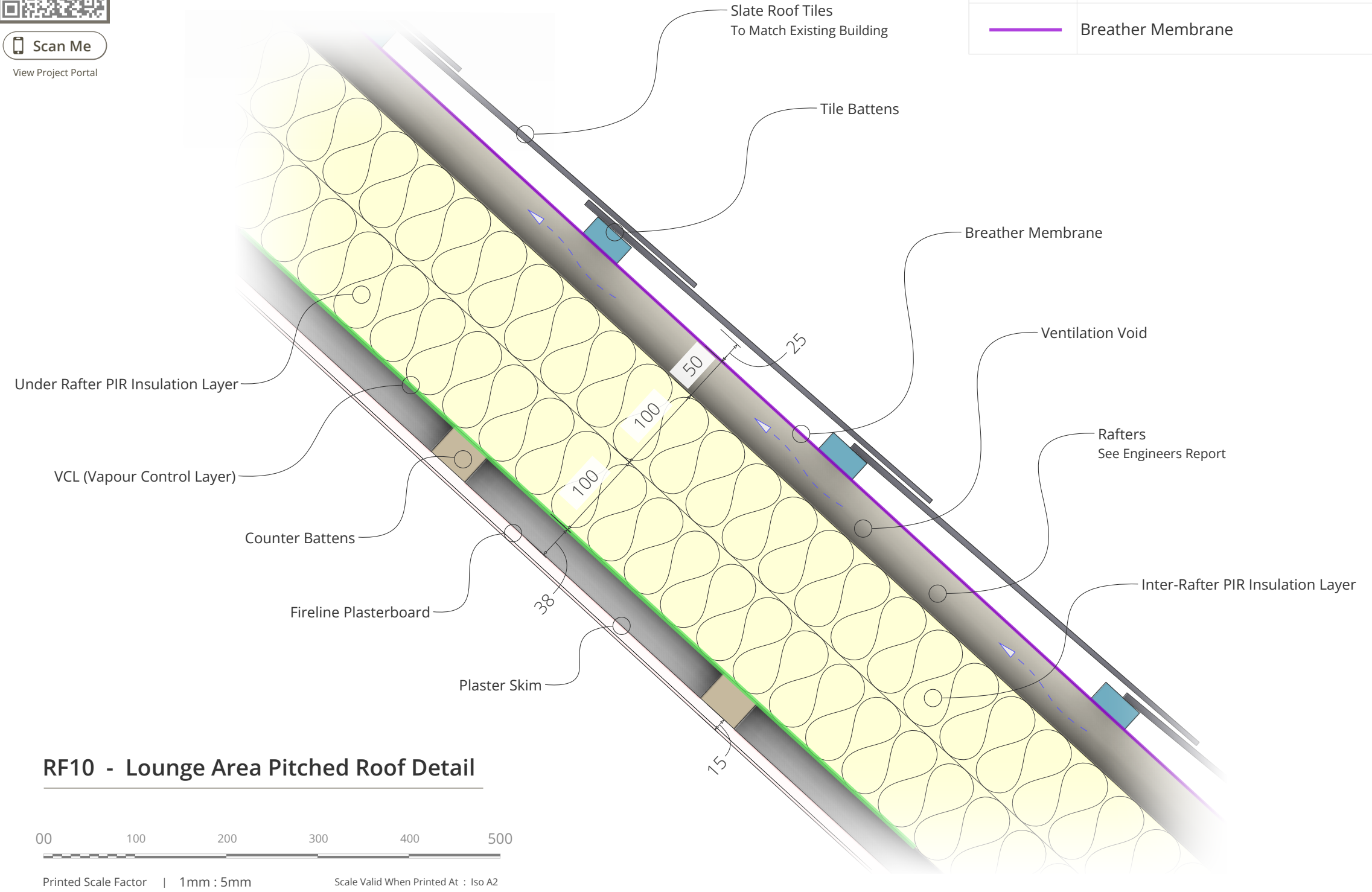
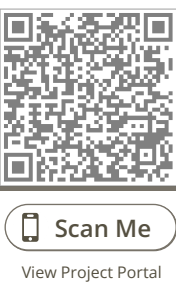




RF10 - Lounge Area Pitched Roof Detail

Calculation			
U = 1 / (0.04 + 0.00 + 0.00 + 0.00 + 0.00 + 3.01 + 4.55 + 0.00 + 0.48 + 0.05 + 0.01 + 0.10) = 1 / 8.24 = 0.12 W/m²K			
Layer	Thickness (mm)	Lambda (W/mK)	R-value (m²K/W)
External Surface Resistance (Rse)	-	-	0.040
Roof Tile Covering - Natural Slate (Double Lap)	7	-	0.000
External Tile Battens - Softwood (50x25mm)	25	-	0.000
Breather Membrane - Tyvek Supro Plus	0.5	-	0.000
Ventilated Air Gap - Within Rafter Zone (BS 5250)	50	-	0.000
Insulation - Inter-Rafter PIR (Bridged 10.4% by Rafters)	100	0.028	3.005
Insulation - Continuous Under-Rafter PIR (GA4000)	100	0.022	4.545
VCL - Virgin Polythene (1000g)	0.25	-	0.000
Service Void - 38mm Batten Cavity (Bridged 12.5%)	38	0.078	0.478
Internal Lining - Gyproc FireLine (12.5mm)	12.5	0.250	0.050
Internal Finish - Thistle Multi-Finish Skim	3	0.570	0.005
Internal Surface Resistance (Rsi)	-	-	0.100
Total R-value			8.223
Result			
U-Value			0.12 W/m²K

Value Compliant - England Part L1B : Roof U-value 0.15 W/m²K.

Element	Specifications
Roof Tile Covering	Remove a Roof slate from the existing building and accurately measure in order to supply matching sized tiles for the extension, supply and install nominal *(508mm x 254mm) 'Countess' Natural Welsh Slate or high-grade Spanish equivalent (Graded T1, S1, A1 to BS EN 12326). Thickness: Nominal 5mm - 7mm. Fix in a double-lap configuration using two 30mm x 3.35mm copper nails per slate. Headlap: 100mm (Verified for 42.5° pitch in exposed zone). Contractor to verify aesthetic match with host dwelling.
External Tile Battens	Fix 50mm x 25mm finished size preservative-treated softwood battens, fully graded and stamped to BS 5534. Fix to rafters using 65mm x 3.35mm galvanised ring shank nails. Note: 50mm width is critical for high wind uplift resistance.
Breather Membrane	Install DuPont Tyvek Supro Plus (or approved high-density equivalent) breather membrane with integral adhesive tape. Lay horizontally across rafters with a slight drape. All laps must be sealed using the integral tape to create a wind-tight barrier to prevent thermal bypass.
Rafters	Construct roof structure using 47mm x 150mm C24 structural grade softwood rafters at 450mm centres (AS PER ENGINEERS REPORT). Ensure timbers are plumb and true on the underside to receive the continuous insulation boards.
Inter Rafter Insulation	Friction fit 100mm Celotex GA4000 (or equivalent PIR) tightly between rafters. Ensure a clear 50mm ventilated air gap is maintained between the top of the insulation and the underside of the breather membrane (BS 5250 compliance). Do not compress insulation.
Under Rafter Insulation	Install a continuous layer of 100mm GA4000 (or equivalent PIR) to the underside of the rafters. Boards must be run perpendicular to rafters. Joints must be tightly butted. This layer eliminates thermal bridging through the timbers.
Vapour Control Layer (VCL)	Supply and install a dedicated 1000-gauge (250 micron) virgin polythene Vapour Control Layer (e.g., Visqueen High Performance VCL) over the warm side of the continuous PIR insulation. This must be installed BEFORE the counter-battens. Lap all joints by minimum 150mm and seal with double-sided butyl tape. Seal perimeter to walls and floor plate with acoustic/airtight sealant. Rationale: Due to the 50/50 split of insulation (100mm in/100mm out), a high-grade VCL is critical to prevent interstitial condensation at the interface.
Counter-Battens	Install 38mm x 50mm treated softwood battens horizontally (perpendicular to rafters) at 400mm vertical centres. These battens clamp the VCL and Insulation to the rafters. The 38mm depth creates the service void for cabling. Warning: Ensure batten thickness accommodates any proposed recessed lighting depth; if installing downlights, use low-profile fire-rated units or install thermal hoods to prevent overheating cables.
Structural Fixings	Fix the Counter-Battens through the VCL and 100mm Under-Rafter PIR into the timber rafters using specialised Structural Insulation Screws (e.g., Timco Insulation Fixings, EJOT Dabo, or Spit). Total fixing length approx 200mm (38mm Batten + 100mm PIR + 60mm Rafter Embedment). Fixings must be driven at a 30-40° upward angle to resist shear load (sliding) of the heavy insulation layer. DO NOT use standard screws.
Internal Lining	Fix 12.5mm Gyproc FireLine (or Knauf Fire Panel) to the horizontal counter-battens. Use standard drywall screws long enough to penetrate the batten but NOT puncture the VCL behind it. Stagger joints.
Internal Finish	Apply 3mm Thistle Multi-Finish skim coat to a smooth finish.
Design Rationale	This High-Performance roof exceeds Part L 2025 standards (Target U-Value <0.13 W/m²K). The 100mm continuous internal layer is utilised to break thermal bridging. A dedicated VCL is positioned behind the service void to protect the airtight line from electrical penetrations. Structural fixings are specified to manage the shear load of the thick insulation on the 42.5° pitch.
Important Considerations	1. LIGHTING: Do not puncture the VCL for light fittings. Use surface mounted lights or ensure recessed fittings fit within the 38mm batten depth without touching the VCL. 2. CONDENSATION: The VCL continuity is absolute; any tears during construction must be repaired with proper tape immediately. 3. LOADS: The insulation is held by the angled screw fixings; the angled driving technique to prevent ceiling slump.
Standards	BS 5534 (Slating/Tiling), BS 5250 (Control of Condensation), BS EN 13165 (PIR Insulation), Part L Vol 1 (2025 Compliance).

Building Control Approval Submission

This drawing is submitted for Building Control approval. Construction must not commence until the Inspector sanctions the proposed structural details, thermal performance, and moisture integrity. The Principal Contractor is fully responsible for coordinating all statutory inspections to verify strict compliance with the Building Regulations 2010.